

(Continued from *NOWELE* vol. 7)

Patrick V. Stiles

THE FATE OF THE NUMERAL '4' IN GERMANIC (3)

3. The ordinal

3.1. Indo-European

In chapter IV of his *Studies in the Indo-European System of Numerals* (1960), Szemerényi demonstrates that the seemingly Proto-Indo-European ordinal suffix *-to-* arose at a relatively late stage through mis-segmentation of IPIE **dekmtó-* '10th'. Compare already Szemerényi 1949:191 fnl: 'as in various territories the final *-t* in **dekmt* disappeared, **dekmtos* was interpreted as containing a suffix *-to* (in others a new ordinal was built in conformity with the preceding numerals, cf. Lat. *decimus*)'; note also 1960:68. Once created in this way, the new suffix *-to-* was able to spread from '10th' to the other ordinals – see especially 1960:92, first table. (This information is repeated in summary form in Szemerényi 1980:209-211.)

The earliest method of forming ordinals in Indo-European was addition of the suffix **-ó-* to the stem of the cardinal, which appeared in zero-grade form; see also Risch 1962:135-136. Indeed, as far as '4th' is concerned, we have already had occasion to observe (§1.6.4.0. REMARK) that the earliest reconstructable form is (unattested) **(k^w)tur-ó-*. This is indirectly reflexed in Vedic *turīya-*, Avestan *tūirya-*, which has been contaminated by '3d'. Sommer (1948) thought he had identified an exact correspondent of this Indo-Iranian form in Hitt. *LÚduianalli-*, supposedly meaning 'fourth (person)' and based on a

Hittite ordinal **tu₄iana-*, enlarged from **tu(r)₄ia-*. This interpretation of the word is disputed, however, cf. Puhvel 1978:99, who suggests that it rather means 'second (person)' and is based on an ordinal **du₄ia-ana* 'second'.

At a later (presumably post-Indo-European) date, a type **k^wet₄ur-to-*, or **k^wetur-to-* (with analogical syllabification), was formed, re-approximated to the cardinal **k^wetu(o)r-* and extended with the new suffix *-tó-* (cf. Szemerényi 1960:79). Examples of the former variant are Gk. (Hom.) τέτρατος, OCS *četrŭgtŭ*, Li. *ketvīrtas*, and also Lat. *quārtus*, from earlier **kwa(t)wortos* by dissimilatory loss of *-t-*; examples of the latter variant are OInd. *caturthā-* (AV+), with secondary *-tha-* (cf. Szemerényi 1960:87 fn97), Latv. *ceturtais*.

3.2. From Indo-European to Proto-Germanic

3.2.0. 'In the Germanic languages', writes Szemerényi (1960:88-89), 'the suffix *-ðā-* spread from "10th" and "5th-6th" to the ordinals between them, in most cases being added directly to the cardinal'. Accordingly, on page 94, he gives the following table of developments, prefixing it thus:

The distinguishing feature of the Germanic set is the generalization of the suffix *-to-* in the series '4th-10th':

<i>*trit(i)yó-</i>		<i>*priðja-</i>
<i>*k^w(e)tur(i)yo-</i>	<i>*k^wetw₄tó-</i>	<i>*feðwurðā- (*fez₄wurðā- ?)</i>
<i>*penk^wto-</i>	}	<i>*fimfta-</i>
<i>*seksto-</i>		<i>*sehsta-</i>
<i>*septm₄mo-</i>	}	<i>*seðundā-</i>
<i>*oktwo-</i>		<i>*ahtōðā-</i>
<i>*newnno-</i>		<i>*newundā-</i>
<i>*dekmtó-</i>	<i>*newntó-</i>	<i>*tehundā-</i>

The Fate of the Numeral '4' in Germanic

Compare also the similar table constructed by Hamp 1952:137-138. (On OHG *sehto*, by-form of *sehsto*, cf. Braune-Eggers 1975:§278 A.1; note also Szemerényi 1960:77-78.)

We may note that Szemerényi's explanation of the origin of the suffix *-to-* (reported above, §3.1.) would date its spread in Germanic later than the loss of final occlusives in polysyllables. This would have chronological implications for the late Indo-European date for the creation of **p(e)nk^w-to-* posited by Szemerényi 1960:86, 87, 92 and suggested by the evidence of the daughter languages. It is possible therefore that *-to-* spread from **dekm̥tó-* by contamination already in Indo-European times. Alternatively and fully in keeping with Szemerényi's basic contention, the appearance of **p(e)nk^w-to-* in the various daughter languages could be the result of parallel post-Indo-European developments.

3.2.1. The Germanic forms of the ordinal '4th' cannot be traced directly to the first of the Germanic pre-forms given in Szemerényi's table. The ordinal is unattested in Gothic, but ON *fjórði*, OE *feorða*, OS *fiorðo* and eOHG *feortho** all point to Proto-Germanic **feurþan-* (so, for example, Kluge 1913:259, Cowgill 1970:147 [n21]). This pre-form indicates that Germanic (like Latin, cf. §3.1. above), has dissimilatory loss of the first *-t-* (or reflex). Obviously, the dissimilation is best motivated if it took place before the operation of Verner's Law, when the sequence was *t ... t* (or *þ ... þ*). (We may also note that PGmc. **feurþan-* is more easily derived from a post-Indo-European shape **k^we(t)urto-*, than from the **k^wetw̥rto-* given by Szemerényi, cf. further §3.2.2.).

As this dissimilation seems such a well motivated development (compare also the parallel of '7th', mentioned below) there is no need to posit an ordinal **fidaurþa**, or the like, for Gothic. Accordingly, I reject the Go. *'*fidurda'* constructed by Szemerényi 1980:210; such a

form need not contravene Thurneysen's Law (of spirant dissimilation in non-initial syllables) if we assume that the Gothic change of **rð > rđ* was earlier. So I would envisage Go. *fiurþa**. If, as seems likely, 'Crimean Gothic' *furdei-* is a mistake for *fiirdei**, it is presumably related to the Germanic ordinal form (so Stearns 1978:135-136). Whether or not it is evidence for (Bible) Gothic is another matter, however, cf. Grønvik 1983a.

It follows from the above that we can dispense altogether with the second, queried, Germanic pre-form **fe₃wurðā-* given by Szemerényi (1960:94), for it is superfluous. Besides, in view of our discussion of the medial consonantism of the cardinal (above §1.3.), the only plausible way to derive it would be by dissimilation from **feðwurðā-* (as suggested by Hamp 1976), and, as we have just seen, there is no direct evidence for the existence of this latter form in Germanic.

REMARK. As implied in Szemerényi's table (and stated by him 1960:35), medial *-t-* has also suffered dissimilatory loss in the Germanic ordinal '7th'. Compare already Kluge 1879b: 393-394, Sievers 1879:575, Brugmann 1895, followed by many others; note also Hamp 1952.

The cardinal '7', PGmc. **seðun*, for expected **seftu*, can be explained as extracted from the ordinal (as can **ne(w)un*, not **ne(w)u*, so Hamp 1952). An alternative is to suppose that the stem-final **-t#* of the cardinal '10' spread to the other cardinals with final syllabic nasal by contamination. This was suggested by Hirt 1932:110; we may note that Seebold 1982:188 cites a pre-form **newmt*, though without further comment. (The obiter dictum of Szemerényi 1960:127 n53, 'Gmc. **neun* **seðun* with *-n* retained after **tehun*, the only numeral where it was legitimate', is similar in spirit if not in substance.) The development would be as follows:

The Fate of the Numeral '4' in Germanic

IE				(1)PGmc.		
*septṛ́n	>	*sep(t)úmt	>	*sepúnt	>	*seḃun
*októu	>	*októu	>		>	*ahtau
*néun	>	*néwunt	>		>	*ne(w)un
*dékm̃t	>	*dékumt	>		>	*tehun

3.2.2. There is a further mismatch between the PGmc. **feurþan-* we have reconstructed on the basis of the Germanic dialects and the pre-Gmc. **k^wetwrtó-* given by Szemerényi (1960:94, cited above §3.2.0.). In showing a voiceless dental, the Germanic form would appear to contradict Verner's Law.

For some reason, '4th' shares this peculiarity with '8th', cf. OHG *ahtodo*, OE *eahtoða*, Go. *ahtudin* (with Thurneysen's Law!).

REMARK 1. Taken in isolation, Go. *ahtudin* (only L 1/59) does superficially appear to reflect a form with earlier *+-tō-*, a fact which has misled many scholars, principally Indo-Europeanists who have not looked beyond this form at the other Germanic evidence. It is noteworthy that Thurneysen did include it as an example of spirant dissimilation in his original article (1898:211), though it is not clear whether he was aware of the issue at hand (cf. 1898:213).

In view of the secondary extension of an **-nþan* ordinal suffix in Ingvaemonic (see §3.3., below), OHG *ahtodo* is the crucial piece of evidence here. Because of assimilation following syncope, ON *áttri* does not give us any relevant information (cf. §3.2.4.).

Of course, Verner's Law cannot be directly contradicted; it is one of the best-established sound laws in the history of Indo-European linguistics (and provided the tertium comparationis for the reconstruction of the position of the Indo-European accent). Rather, the pre-Germanic forms for '4th' and '8th' given by Szemerényi (and many others) are in need of revision as regards the position of the accent. (From this point

of view, both of the Proto-Germanic forms given by Szemerényi 1960:94, and also the Go. **fidurda* he constructs 1980:210 - mentioned above §3.2.1. - are to be rejected.)

As Szemerényi is at such pains to argue throughout chapter IV of his book (1960) that the *-to*-ordinals are post-Indo-European, it is perhaps not surprising that their accent should be 'un-Indo-European'. Rather, this fact supports Szemerényi's contention. It is certainly odd that the place of the accent should be 'irregular' in precisely '4th' and '8th'. (The fact that one is the other doubled may or may not be significant; it is interesting to note that Szemerényi 1960:85-86, 87, 100 suggests that the new ordinal suffix **-to-* spread from '10th' first to '5th'; also 1980:209. For an attempted explanation of why these two ordinals alone should have this 'un-Indo-European' accentuation, see §3.2.3. below.)

Verner himself was well aware of the problem of the voiceless spirant in '4th' (he does not mention '8th', doubtless for the reason mentioned above, §3.2.2. REMARK 1; Thurneysen's Law was not 'promulgated' until over twenty years later, Wrede 1897:177-178, Thurneysen 1898). In his celebrated article, Verner noted that the Germanic form 'stimmt nicht mit altind. *caturthá*-', and suggested: 'möglicherweise ist aber die betonung im germ. *fedvōrþan*- in übereinstimmung mit der betonung der cardinalzahl gewesen; vgl. lit. *ketvìrta*-s, russ. *četyrtyj*, bulg. *četrŭti*' (1877:118).

Though Verner's explanation provides an excellent motivation for the position of the accent, the pre-form he gives needs some adjustment. If we assume that the new *-to-* ordinal was built on the zero-grade stem-form of the cardinal (which retained its accent), we can account for every feature of the Germanic reflex. With dissimilatory loss of the first *-t-* and the adoption of weak adjective inflection, **k^wetúrto-* leads (via c. **péurto-*) to the **feurþan-* we reconstructed above on the basis of the (North-West) Germanic evidence. For '8th', similarly: **oktōu-* + *-to-*, yielding PGmc. **ahtaupan-*. As indicated in

The Fate of the Numeral '4' in Germanic

these reconstructions, the rebuilding must have preceded the operation of Verner's Law, otherwise we would not obtain the attested voiceless dental.

REMARK 2. We may additionally note that **dekmtó-*, the pre-Germanic form for '10th' given by Szemerényi in his table (cited above §3.2.0.), should yield PGmc. **teʒunda-*, not the **tehunda-* he cites. Hamp was aware of this problem (1952:137), and suggested that the ordinal has been rebuilt on the cardinal (1952:138-139).

Hamp made no mention of the following Ingvaeonic material, however: Old English (Anglian) *teogeða* etc. (Sievers-Brunner 1965: §328), OFris. *tegotha* (Steller 1928: §79; cf. also §3.3.4., below), and OS *tegotho*, *tegatho* (Holthausen 1921: §388). The voiced guttural found in these forms certainly needs to be accounted for. In view of the geographical restriction, it seems to me more likely that these forms are a secondary Ingvaeonic development, than that they reflect the expected Verner's Law outcomes. The most obvious source is the voiced guttural which appears in '9' and 9th' in these dialects and only there - on this, cf. most recently Seebold 1982:188, Markey 1984. (It has usually been held that a similar contamination has produced OFris. *si(u)gun* etc., cf. Siebs 1901:1197, Frings et al. 1962:50, Nr.3, but Markey has recently presented powerful arguments for regarding the development as phonological in this numeral, 1984).

This interpretation allows us to take the by-forms OE *teopa*, OFris. *tianda*, OS *tehando*, as correspondents of the (rebuilt) IPGmc. **tehunda-* indicated by Go. *taihunda* (not susceptible to Thurneysen's Law, as *d* was a stop after nasals, Braune-Ebbinghaus 1981: §72), ON *tíondi* and OHG *zehanto*. As shown by the voicing of the suffix in these forms, the re-building of '10th' on the cardinal is later than Verner's Law. Specifically, it was Verner's Law, which created the motivation for the analogy: pre-Verner's Law (**tehun:*) **tehunpa-*; post-Verner's Law (**tehun:*) **teʒunda-* → **tehunda-*.

3.2.3. The evidence discussed here suggests a chronology for the pre-Germanic developments slightly different from that depicted in Szemerényi's table (1960:94, cited in §3.2.0.).

If we accept Szemerényi's motivation (cf. §3.1.), the spread of **-to-* in (pre-) Germanic is to be dated later than the loss of final occlusives in polysyllables. (Otherwise we might take it as a late Indo-European phenomenon, as outlined in §3.2.0.) The unshifted **-t-* in '5th' and '6th' indicates that these were formed earlier than the operation of Grimm's Law. By a similar argument, the dual treatment of the shifted **-t-* shows that Germanic '4th', '7th', '8th', and '9th' must all have been formed earlier than Verner's Law, which was phonologized by the accent shift. (Indeed, the subsequent history of the suffix for the lower ordinals in Germanic is to a large extent the levelling of the variation created by Grimm's and Verner's Laws, see §3.3.)

The fact that '7th' and '9th' reflect the 'Indo-European' accent position, presumably means that they were formed earlier than '4th' and '8th'. One would certainly like to be able to produced a reason as to why this might be so. It seems most natural to try and connect this early spread of **-to-* with the fact that, like '10th' (IPIE **deḱmtó-*), the late Indo-European forms of '7th' (**septmómó-*) and '9th' (**neunnómó-*) contained a syllabic nasal. Alternatively, we could postulate that the early rebuilding of the ordinals '7th' and '9th' was induced by the formation of new cardinals '7' and '9' with final **-t#*, modelled on '10', as discussed above §3.2.1. REMARK. (In this case, the rebuilding of the ordinals on these new cardinals would precede the loss of final **-t#*; this could thus have contributed to the spread of the ordinal suffix **-to-* in Germanic.) As we have already observed, the rebuilding of '10th' is later than Verner's Law (§3.2.2. REMARK).

With reference to Hamp's discussion (1952:138-139) of the most economical way to account for the series of changes reflected in the Germanic set of cardinals and ordinals, we may note the following

The Fate of the Numeral '4' in Germanic

considerations. If we were to accept that the final nasal of the cardinals **seḁun* and **ne(w)un* had indeed been preserved by an earlier final **-t#* (of analogical origin), then we need only operate with a single (derivational-cum-) analogical principle from earliest Proto-Indo-European to post-Verner's Law Germanic: at every stage the ordinal can be regarded as (derived from/) rebuilt on the cardinal. I see no particular virtue in this theoretical rigour, however. In a close-knit system such as that of cardinals and ordinals, analogical influences can go in either direction (cf. also Markey 1984:68); examples of a cardinal extracted (backformed) from the ordinal are to be found in OHG *feor* (cf. §1.3.2.), and Rök *tualf* (cf. §1.4.6.4.).

3.2.4. A summary of the development of the ordinals '3d' to '10th' to late Proto-Germanic can be conveniently presented in the form of a table:

IIE	pre-Gmc. I	pre-Gmc. II	post VL	IPGmc.
<i>*trit(i)ió-</i>				<i>*priðja-</i>
<i>*k^w(e)turó-</i>				<i>*feurþa-</i>
<i>*p(e)nk^w(t)ó-</i>	<i>*penk^wto-</i>	<i>*k^we(t)úrto-</i>		<i>*fimfta-</i>
<i>*ksu(e)ksó-</i>	<i>*seksto-</i>			<i>*sehsta-</i>
<i>*septm^omó-</i>	<i>*septumtó-</i>			<i>*seḁunda-</i>
<i>*okt^ouó-</i>		<i>*okt^outo-</i>		<i>*ahtauþa-</i>
<i>*neunnó-</i>	<i>*newuntó-</i>			<i>*newunḁa-</i>
<i>*dekm^otó-</i>				<i>*teʒunḁa- → *tehunda-</i>

On the Indo-European form of '6', cf. de Saussure 1922:435-439 [= 1892], more recently Hamp 1978, without reference to de Saussure's earlier discussion. (As correctly reported by Hamp, Ernout-Meillet 1959:621 remark: 'Comme l'a vu F. de Saussure, l'initiale de ce nom de nombre a dû être complexe en indo-européen'. Unfortunately,

however, they give no specific reference to de Saussure's discussion, and fail to represent his findings.)

For the sake of simplicity, I operate with the late Proto-Indo-European pre-form for '7th' given here; for further information, see Cowgill 1970:146 [n18] in the first instance.

On the question of the assimilation of *-mt-* to *-nǵ-*, compare the remarks of Hamp 1952:139.

The medial vocalism of Go. *ahtudin* is problematic. Szemerényi 1960:88-89 regards the *-u-* of *ahtudin* as a variant-spelling for 'expected' *-ō-* (also considered possible by Hollifield 1979:56). On morphological grounds (cf. §3.2.2.), I would expect *-au-*, however. I wonder, therefore, whether we might not have here an example of <u> representing etymological *au*. (Aside from the well-known fluctuation in the spelling of the final vowel of the singular case-forms of the *u*-stems, I know of only *uſtō* Mt 27/64, beside usual *aufstō*). Hamp 1952:139 seems to be suggesting that the Go. *-u-* is analogical on **sib-u-nda*, **new-u-nda*, **teh-u-nda* (so van Helten 1905-1906:125). All in all, I think this single attestation is an uncertain basis for firm conclusions.

We may note that ON *átti* is anomalous in having syncope of earlier long **ō*. Rosenfeld 1956-1957:197 and Frings et al. 1962:51 no. 12 express the view that it is modelled on *sétti*.

The medial *-o-* of OHG *ahtōdo* is circumflexed in six instances out of thirty-two (including one of *úberáhtodo*) in Notker's works; see *Althochdeutsches Wörterbuch* for the examples. Professor James C. King has kindly confirmed these readings for me in a letter of 8th July 1985. Unfortunately, the word does not occur in the *Consolatio*, in which the original accentuation is best preserved. I take this fluctu-

The Fate of the Numeral '4' in Germanic

ation to reflect phonetic (rather than phonemic) shortening of a long vowel. This can, of course, reflect either PGmc. *-ō-* or *-au-*.

If, as I believe, OE *eahtoþa* contains the reflex of an earlier long vowel, it regularly lacks syncope and is not rebuilt as Sievers thought (1878:73).

3.3. A note on the voicing of the ordinal suffix in dialectal Germanic

3.3.0. The (late) Proto-Germanic ordinals '4th-10th' reconstructed above (§3.2.4.) can clearly be classed into three groups on the basis of their suffix-form: A '5th' and '6th', with unshifted **-t-*; B '4th' and '8th', with **-þ-*; C '7th', '9th', and '10th', with the sequence **-und-*.

3.3.1. The Old High German reflexes are the product of regular sound-change alone. Nonetheless, the result has been that groups A and C have acquired the same consonant (*-t-*). Group B remained distinct longer: *vierde* and *ahtode* are still found in Middle High German, before being replaced by *vierte* and *ahte*.

3.3.2. Gothic too shows regular reflexes in those ordinals it attests (cf. Braune-Ebbinghaus 1981:§146), the only complication being the operation of Thurneysen's Law in *ahtudin*. This, and the lack of an instance of 'fourth', means that group B cannot be identified on the evidence of Gothic alone (cf. §3.2.2. REMARK 1).

3.3.3. In North Germanic, the merger of Proto-Germanic intervocalic **-þ-* and **-ð-* has removed the distinction between '3d' and '4th'. Group B has ceased to exist insofar as '8th' has been rebuilt. Two new forms are found: *átti* (most probably modelled on *sétti*, as mentioned in §3.2.2. REMARK), and (more recent) *óttundi*, *áttandi* (the result of interference from the surrounding ordinals, *si(a)undi*, *níondi*, *tíonde*, group C).

3.3.4. An analogous rebuilding of '8th' under the influence of group C is found in Ingvaeonic also, compare OFr. *achtunda*, MLG *achtende*. The most notable feature of the Ingvaeonic languages, however, is the occurrence (alongside the inherited shapes) of forms of '7th', '9th', and '10th' (group C) with voiceless dental (as if) from **seþuþo*, **nigupþo*, **teguþo*. These must owe their origin to influence from group B (as suggested by Cowgill 1970:147 [n21]). As in the case of the forms of '10th' with *-g-* (cf. above §3.2.2. REMARK 2), the geographical restriction of the phenomenon suggests that it is not to be put down to seemingly random 'Verner's Law variation' reflecting capricious accentuation in pre-Germanic (the line taken by Siebs 1901:1363, Kluge 1913:259, Kieckers 1928:182, Prokosch 1939:292 – with at least a reasonable supposed motivation – and Rosenfeld 1956-1957:196, among others).

Thus, beside the 'pristine' set of Old High German, Old Saxon and Old English have such forms as: *siuotho*, *niguda*, *tegotho*; *seofopa*, *nigopa*, *teo(ge)pa* (cf. Kluge 1879a:132-133, van Helten 1895:191, Gallée-Lochner 1910:§360, Holthausen 1921:§388). Later-attested Old Frisian only has such a form for '10th', *tegotha*, *tegetha* (Steller 1928:§79.2), and has no counterpart of OS *ahotoþo*, but the forms *achtunda*, *achtenda*, and *achta* (after *fifta* according to Rosenfeld 1956-1957:197 and Frings et al. 1962:51 Nr.12, though Norse, or mediaeval German, influence also seems a possibility; van Helten 1890:§239 regards it as a regularly syncopated form, but I think this doubtful because the medial vowel was originally long). Group C has otherwise gained the upper hand in Old Frisian. To judge by the spelling, OFr. *fiarde* has analogically adopted the *-d-* of group C.

As pointed out to me by Rolf Bremmer, *tegotha* (found only in Old East Frisian) preserves its form in the marginal sense of 'tithe, tax' (cf. the derivative *tegothere* 'tithe-collector') and clearly owes its continued existence to this sense-differentiation; in the primary sense '10th' *tianda*, *tienda* is used. If the explanation of the voiceless dental

The Fate of the Numeral '4' in Germanic

in such forms given above is correct, however, the previous existence of an **ahþo* form of '8th' is presupposed by *tegotha* (so van Helten 1895:191).

The circumstance that we have to do with an analogical replacement **tehun-dan-* → **tehun-þan-* etc. allows us to give a chronological specification to our dialectal determination. The replacement must have preceded the Ingvaenonic loss of nasals before homorganic spirants, and was therefore early.

4. Summary of major conclusions

a) The initial *f-* is most probably the result of an unconditioned phonological development (§1.1.).

b) The cardinal came down into Germanic inflected for case, but not gender. The stem had two allomorphs, **feðwōr* in the endingless nominative-accusative (the phonological reflex of the Indo-European neuter form, §1.2.), and **feður-* in the genitive and dative (which took the endings of the consonant stems), §1.6.

c) The dative was contaminated by the (*i*-stem) numeral '3' already in Proto-Germanic (§1.6.).

d) The late Proto-Germanic inflection was subsequently remodelled in each branch of Germanic. Gothic generalized the stem-form **feðwōr(-)*, but otherwise essentially retained the Proto-Germanic state of affairs. North-West Germanic evidently rebuilt the whole paradigm according to the *i*-stem pattern and created a gender distinction between neuter and masculine/feminine in the nominative-accusative. North Germanic subsequently generalized the stem-form **feður-* and adopted the endings of the strong adjectives. West

Patrick V. Stiles

Germanic generalized the stem-form **feðwōr(-)*, independently of Gothic, and retained the *i*-stem inflection (§1.6.).

e) In West Germanic the medial **-ðw* became **-ww-* as a phonological development (§1.3.).

f) The Proto-Germanic combining-form was inherited **feþur-* (cf. OInd. *cátur-*); this was later prone to replacement by forms influenced to a greater or lesser extent by the cardinal (cf. §2).

g) The ordinal PGmc. **feurþan-* has dissimilatory loss of the first *-t-* of post-IE **k^wetúrto-* (rebuilt on the zero-grade form of the cardinal), cf. §3.

Gayre Institute

University of Edinburgh

GB-Edinburgh EH8 9LW

Acknowledgements: I would like to express my gratitude to the following scholars for having improved earlier drafts of this article: Hreinn Benediktsson, Rolf Bremmer Jr., Thea Bynon, Warren Cowgill, Ottar Grønvik, Terry Hoad, D.R. McLintock. I would also like to thank Nils Århammar, Ernst A. Ebbinghaus, and James C. King for their help on specific points.

Bibliography

- Althochdeutsches Wörterbuch*. 1952-. Eds. E. Karg-Gasterstädt, Th. Frings et al. Berlin.
- Bammesberger, Alfred. 1979. *Beiträge zu einem etymologischen Wörterbuch des Altenglischen*. Heidelberg.
- Bazell, C.E. 1977. 'On the margins of sound-change and analogy'. *CFS* 31:37-45.
- Behaghel, Otto. 1917. 'Zum gotischen Zahlwort'. *PBB* 42:156-158.
- Behaghel, Otto. 1923. *Deutsche Syntax I*. Heidelberg.
- Benediktsson, Hreinn. 1963. 'Some Aspects of Nordic Umlaut and Breaking'. *Lg* 39:409-431.
- Benediktsson, Hreinn. 1980. 'Nordic Umlaut and Breaking: Thirty Years of Research'. Pre-prints of Fourth International Conference of Nordic and General Linguistics. Abstracts (Supplement). Oslo. 10-43.
- Benediktsson, Hreinn. 1982. 'Nordic Umlaut and Breaking: Thirty Years of Research (1951-1980)'. *NJL* 5:1-60.
- Bennett, William H. 1969. 'Pre-Germanic /p/ for Indo-European /k^w/'. *Lg*. 45:243-247.
- Bethge, R. 1900. 'Deklination des Urgermanischen', in Ferdinand Dieter, *Laut- und Formenlehre der altgermanischen Dialekte*. Leipzig. 515-567.
- Boer, R.C. 1920. *Oudnoorsch Handboek* (= Oudgermaansche Handboeken 2). Haarlem.
- Boer, R.C. 1921. 'Gotisch rz'. *Neophil.* 6:228-235.
- Braune, Wilhelm. 1876. 'Über die Quantität der althochdeutschen Endsilben'. *PBB* 2:125-167.
- Braune, Wilhelm and Ernst A. Ebbinghaus. 1981. *Gotische Grammatik*. 19. Auflage. Tübingen.
- Braune, Wilhelm and Hans Eggers. 1975. *Althochdeutsche Grammatik*. 13. Auflage. Tübingen.

- Brøndum-Nielsen, Johs. 1928-1973. *Gammeldansk grammatik i sproghistorisk fremstilling*. 8 vols. København.
- Brown, T.J., F. Wormald, A.S.C. Ross, E.G. Stanley (eds.). 1969. *The Durham Ritual*. (Early English Manuscripts in Facsimile 16). Copenhagen.
- Brugmann, Karl. 1890. 'Zur Bildung des nom. acc. plur. neut.', in Hermann Osthoff and K.B. *Morphologische Untersuchungen*. V. Leipzig. 52-61.
- Brugmann, Karl. 1895. 'Ahd. sibun und āband'. *IF* 5: 376-379.
- Brugmann, Karl 1904. *Kurze vergleichende Grammatik der indogermanischen Sprachen*. Strassburg.
- Bugge, Sophus. 1873-1895. 'Tolkning af runeindskriften på Rökstenen i Östergötland'. *Antiqvarisk Tidskrift för Sverige* 5:1-148. [1873-1878].
- Buma, W.J., W. Ebel, M. Trachter-Schubert (eds.). 1977. *Jus Municipium Frisonum*. (Altfriesische Rechtsquellen 6). 2 vols. Göttingen.
- Campbell, A. 1959. *Old English Grammar*. Oxford.
- Cowgill, Warren. 1970. 'Italic and Celtic Superlatives and the Dialects of Indo-European', in *Indo-European and Indo-Europeans*, ed. George Cardona et al. Philadelphia. 113-153.
- Cowgill, Warren. 1980. 'The etymology of Irish *guidid* and the outcome of *g^{wh} in Celtic', in *Lautgeschichte und Etymologie*, Akten der VI. Fachtagung der Indogermanischen Gesellschaft, ed. Manfred Mayrhofer, Martin Peters, Oskar E. Pfeiffer. Wiesbaden. 49-78.
- Cubbin, G.P. 1979. 'A case of Homonymic Clash in Germanic'. *IF* 84:226-236.
- Ernout A. and Antoine Meillet. 1959. *Dictionnaire étymologique de la langue latine. Histoire des mots*. 4e édition. Paris.
- Frings, Theodor, Gotthard Lerchner and Gertraud Müller. 1962. 'Ingwäonisches in den Bezeichnungen der Zehnerzahlen'. *PBB* (H) 84:1-66.

The Fate of the Numeral '4' in Germanic

- Gallée, Johan Hendrik and Johannes Lochner. 1910. *Altsächsische Grammatik*. Göttingen.
- Girvan, Ritchie. 1931. *Angelsaksisch Handboek*. (= Oudgermaansche Handboeken 4). Haarlem.
- Grønvik, Ottar. 1983a. *Die dialektgeographische Stellung des Krimgotischen und die krimgotische cantilena*. Oslo.
- Grønvik, Ottar. 1983b. 'Runeinnskriften på Rök-stenen'. *MoM* 101-149.
- Hamp, Eric P. 1952. 'The Anomaly of Germanic "7"'. *Word* 8:136-139.
- Hamp, Eric P. 1976. 'Etymologies: OE *feower*, OHG *niun*'. *MGS* 1:1-2.
- Hamp, Eric P. 1978. 'Indo-European "6"', in *Festschrift for Archibald A. Hill*, ed. Mohammad Ali Jazayery et al. The Hague. III.81-90.
- Hamp, Eric P. 1980. 'IE. **()kuon* - "dog"'. *IF* 85: 35-42.
- Harding, Erik. 1942. 'Got. *fidwōr*: ags. *fēower*: isl. *fiogor* osv.'. *Språkvetenskapliga problem i ny belysning* §75 (Part 5:38-40). Lund.
- Harding, Erik. 1944. 'Om räkneordet "fyra" än en gång'. *Språkvetenskapliga problem i ny belysning* §85 (Part 6:23-25). Lund.
- Hartmann, F. 1900. 'Konsonantismus des Althochdeutschen', in *Laut- und Formenlehre der altgermanischen Dialekte*, ed. Ferdinand Dieter. Leipzig. 293-343.
- Hellquist, Elof. 1939. *Svensk etymologisk ordbok*. Ny upplaga. Lund.
- Helten, W.L. van. 1890. *Altostfriesische Grammatik*. Leeuwarden.
- Helten, W.L. van. 1895. 'Zur altsächsischen Grammatik'. *IF* 5:182-193.
- Helten, W.L. van. 1900. 'Zu den malbergischen glossen und den salfränkischen formeln und lehnwörtern in der Lex Salica'. *PBB* 25:225-542.
- Helten, W.L. van. 1905-1906. 'Zum germanischen Zahlwort'. *IF* 18:84-126.
- Heusler, Andreas. 1932. *Altisländisches Elementarbuch*. 3. Auflage. Heidelberg.
- Hirt, Hermann. 1932. *Handbuch des Urgermanischen*. Teil II: Stammbildungs- und flexionslehre. Heidelberg.

- Hollifield, Patrick Henry. 1979. 'Final * $\bar{o}$ in Monosyllables in North and West Germanic'. *Die Sprache* 25:54-56.
- Holthausen, Ferdinand. 1921. *Altsächsisches Elementarbuch*. Heidelberg.
- Ilkow, Peter. 1968. *Die Nominalkomposita der altsächsischen Bibeldichtung*. Göttingen.
- Jansson, Valter. 1936. 'Räkneordet "fyra"', in *Bidrag till Nordisk Filologi tillägnade Emil Olson*. Lund. 272-287.
- Kieckers, Ernst. 1928. *Handbuch der vergleichenden gotischen Grammatik*. München.
- Klein, Thomas. 1977. *Studien zur Wechselbeziehung zwischen altsächsischem und althochdeutschem Schreibwesen und ihrer sprach- und kulturgeschichtlichen Bedeutung*. Göttingen.
- Kluge, Friedrich. 1879a. *Beiträge zur Geschichte der germanischen Conjugation*. (Quellen und Forschungen zur Sprach- und Kulturgeschichte der germanischen Völker 32). Strassburg.
- Kluge, Friedrich. 1879b. 'Grammatisches. (IV. Zu den Numeralien)'. *PBB* 6:393-399.
- Kluge, Friedrich. 1882. 'Sprachhistorische miscellen'. *PBB* 8:506-539.
- Kluge, Friedrich. 1886. 'Zur altgermanischen Sprachgeschichte'. *PBB* 11:557-562.
- Kluge, Friedrich. 1908-1909. 'Gotische Miszellen'. *ZDW* 10: 64-65.
- Kluge, Friedrich. 1913. *Urgermanisch. Vorgeschichte der altgermanischen Dialekte*. 3. Auflage. Strassburg.
- Kock, Axel. 1895. 'Zur behandlung des durch u entstandenen brechungsdiphthongs in den altnord. sprachen'. *PBB* 20:117-140.
- Kock, Axel. 1898. 'Studier i de nordiska språkens historia'. *ANF* 14:213-270.
- Kock, Axel. 1917. 'En fornnorsk och östnordisk ljudlag'. *ANF* 33:241-275.
- Kögel, Rudolf. 1884. 'Über w und j im westgermanischen'. *PBB* 9:523-544.

The Fate of the Numeral '4' in Germanic

- Krahe, Hans. 1961. 'Altgermanische Kleinigkeiten'. *IF* 66:35-43.
- Krahe, Hans and Wolfgang Meid. 1969. *Germanische Sprachwissenschaft*. II Formenlehre. Berlin.
- Krahe, Hans and Elmar Seebold. 1967. *Historische Laut- und Formenlehre des Gothischen*. Heidelberg.
- Krause, Wolfgang. 1948. *Abriss der altwestnordischen Grammatik*. Halle/Saale.
- Krause, Wolfgang. 1968. *Handbuch des Gotischen*. 3. Auflage. München.
- Lejeune, Michel. 1970. 'Phonologie osque et graphie grecque'. *REA* 72:271-316.
- Leskien, A. 1876. *Die Declination im Slavisch-Litauischen und Germanischen*. Leipzig.
- Loewe, Richard. 1902. 'Ostgermanisch-westgermanische neuerungen bei zahlwörtern'. *PBB* 27:75-88.
- Markey, Thomas L. 1976. *Germanic Dialect Grouping and the Position of Ingvæonic*. (Innsbrucker Beiträge zur Sprachwissenschaft 15). Innsbruck.
- Markey, T.L. 1984. 'IE -w- > Gmc. -g- and OFr. *si(u)gun* "7" and *ni(u)gun* "9"', in *Miscellanea Frisica*, ed. N.R. Århammer et al. Assen. 67-77.
- Martinet, André. 1972. 'Dès labio-vélaires aux labiales dans les dialectes indo-européens', in *Indo-Celtica. Gedächtnisschrift für Alf Sommerfelt* (Comentationes Societatis Linguisticae Europaeae 2), ed. H. Pilch und Joachim Thurow. München. 83-93.
- Meillet, Antoine. 1929. Review of Kieckers 1928. *BSL* 29/2:170-172.
- Meillet, Antoine. 1937. *Introduction à l'étude comparative des langues indo-européennes*. 8e édition. Paris.
- Microfiche Concordance to Old English*, A. 1980. Ed. R.L. Venezky and A. DiPaolo Healey. Newark/Toronto.
- Mossé, Fernand. 1956. *Manuel de la langue Gotique*. 2e édition. Paris.

- Nielsen, Hans F. 1981. *Old English and the Continental Germanic Languages. A Survey of Morphological and Phonological Interrelations*. (Innsbrucker Beiträge zur Sprachwissenschaft 33). Innsbruck.
- Noreen, Adolf. 1890. 'Några fornnordiska judlagar'. ANF 6:303-339.
- Noreen, Adolf. 1892. *Altnordische Grammatik. I Altisländische und altnorwegische Grammatik*. 2. Auflage. Halle/Saale.
- Noreen, Adolf. 1897. 'Svenska etymologier', in *Skrifter utg. af k. hum. Vetenskapssamfundet i Upsala* 5.
- Noreen, Adolf. 1903. *Altnordische Grammatik. I Altisländische und altnorwegische Grammatik*. 3. Auflage. Halle/Saale.
- Noreen, Adolf. 1904. *Altnordische Grammatik. II. Altschwedische Grammatik*. Halle/Saale.
- Noreen, Adolf. 1913. *Geschichte der nordischen Sprachen, besonders in altnordischer Zeit*. 3. Auflage. Strassburg.
- Noreen, Adolf. 1923. *Altnordische Grammatik. I Altisländische und altnorwegische Grammatik*. 4. Auflage. Halle/Saale.
- Osthoff, Hermann. 1878. 'Formassociation bei zahlwörtern', in H.O. and Karl Brugmann *Morphologische Untersuchungen*. I. Leipzig. 92-103.
- Paul, Hermann. 1894. 'Gotisch *ai* vor *r*'. IF 4:334-335.
- Pokorny, Julius. 1959. *Indogermanisches etymologisches Wörterbuch*. Vol. I. Bern.
- Prokosch, E. 1939. *A Comparative Germanic Grammar*. Philadelphia.
- Puhvel, Jaan. 1978. 'Remarks on "two" in Hittite'. KZ 92:98-107. (Reprinted in Jaan Puhvel, *Analecta Indoeuropaea: delectus operum minorum plerumque Anglice aliquando Francogallice editorum annos 1952-1977 complectens* = IBS 35.)
- Richthofen, Karl Freiherr von. 1840. *Friesische Rechtsquellen*. Berlin. (Reprinted in two volumes Aalen 1960-1961.)
- Risch, Ernst. 1962. 'Das indogermanische Wort für "hundert"'. IF 67:129-141.

The Fate of the Numeral '4' in Germanic

- Rosenfeld, Hans-Friedrich. 1956-1957. 'Die germanischen Zahlen von 70-90 und die Entwicklung des Aufbaus der germanischen Zahlwörter'. *Wiss Zs. d. U. Greisfw.* 6:171-217.
- Ross, A.S.C. 1954. 'Contribution to the study of *u*-flexion (concluded)'. *TPS* 85-128.
- Ross, A.S.C. 1974. 'Anglo-Saxon "e" to "i" before "u"'. *N&Q* 219:123.
- Saussure, Ferdinand de. 1922. *Reçueil des publications scientifiques*. Genève. (Re-issued 1984.)
- Scherer, Wilhelm. 1868. *Zur Geschichte der deutschen Sprache*. Berlin.
- Schlyter, C.J. 1841. ed. *Westmannalagen*. (Samling af Sveriges gamla lagar 5). Lund.
- Schmidt, Johannes. 1870. Review of Leo Meyer, *Die gotische Sprache* (Berlin 1869). *KZ* 19:268-296.
- Schmidt, Johannes. 1880. Review of Kluge 1879. *AfdA* 6:117-129.
- Schmidt, Johannes. 1889. *Die Pluralbildungen der indogermanischen Neutra*. Weimar.
- Schulze, Wilhelm. 1934. *Kleine Schriften*. Göttingen.
- Schwarz, E. 1951. *Goten, Nordgermanen, Angelsachsen*. Bern-München.
- Seebold, Elmar. 1967. 'Die Vertretung von idg. *g^h* im Germanischen'. *KZ* 81:104-133.
- Seebold, Elmar. 1980. 'Etymologie und Lautgesetz', in *Lautgeschichte und Etymologie*, Akten der VI. Fachtagung der Indogermanischen Gesellschaft, ed. Manfred Mayrhofer, Martin Peters, Oskar E. Pfeiffer. Wiesbaden. 431-484.
- Seebold, Elmar. 1982. 'Der Übergang von idg. -w- zu germ. -k- und -g-'. *IF* 87:172-194.
- Siebs, Theodor. 1901. 'Geschichte der friesischen Sprache', in *Grundriss der germanischen Philologie*, ed. Hermann Paul. 2. Auflage. Strassburg. I. 1152-1464.
- Sievers, Eduard. 1878. 'Zur accent- und lautlehre der germanischen sprachen. II, III'. *PBB* 5:63-163.

- Sievers, Eduard. 1879. 'Kleine beiträge zur deutschen grammatik. VII. Varia. 4) Zu den numeralien'. *PBB* 6:575-576.
- Sievers, Eduard and Karl Brunner. 1965. *Altenglische Grammatik nach der angelsächsischen Grammatik von Eduard Sievers*. 3. Auflage, von K. Brunner. Tübingen.
- Sipma, P. 1927-1941. *Oudfriesche Oorkonden*. I-III. (Oudfriesche Taalen Rechtsbronnen I-III). Den Haag.
- Sjölin, Bo. 1969. *Einführung in das Friesische*. Stuttgart.
- Smith, A.H. 1956. *English Place-Name Elements*. Part One. (EPNS 25). Cambridge.
- Sommer, Ferdinand. 1949. 'Hethitisch $L\ddot{U}$ duḫanalli-'. *IF* 59:205-207.
- Stang, Christian S. 1966. *Vergleichende Grammatik der Baltischen Sprachen*. Oslo.
- Stearns, Jr., MacDonald. 1978. *Crimean Gothic. Analysis and Etymology of the Corpus*. (Studia linguistica et philologica 6). Saratoga.
- Steller, Walther. 1928. *Abriss der altfriesischen Grammatik*. Halle/Saale.
- Szemerényi, O.J.L. 1949. Review of Emile Benveniste, *Noms d'agent et noms d'action en indo-européen* (Paris 1948). *ArchL* 1:187-191.
- Szemerényi, O.J.L. 1960. *Studies in the Indo-European System of Numerals*. Heidelberg.
- Szemerényi, O.J.L. 1980. *Einführung in die vergleichende Sprachwissenschaft*. 2. überarbeitete Auflage. Darmstadt.
- Thurneysen, Rudolf. 1898. 'Spirantenwechsel im Gotischen'. *IF* 8:208-214.
- Trautmann, R. 1912. Review of R. Loewe, *Germanische Sprachwissenschaft* 2. Auflage (Leipzig 1911), and *Deutsches Wörterbuch* (Leipzig 1910). *AfdA* 35:105-109.
- Verner, Karl. 1877. 'Eine ausnahme der ersten lautverschiebung'. *KZ* 23:97-130.
- Vetter, Emil. 1953. *Handbuch der italischen Dialekte*. I. Band. Heidelberg.

The Fate of the Numeral '4' in Germanic

- Wackernagel, Jakob. 1881. 'Zum zahlwort'. *KZ* 25:260-291.
- Wackernagel, Jakob and Albert Debrunner. 1930. *Altindische Grammatik*. III. Band Nominalflexion-Zahlwort-Pronomen. Göttingen.
- Wilmanns, W. 1909. *Deutsche Grammatik*. Dritte Abteilung: Flexion. 2. Hälfte: Nomen und Pronomen. Strassburg.
- Wrede, Ferdinand. 1897. Review of Wilhelm Streitberg, *Urgermanische Grammatik* (Heidelberg 1896). *DLZ* 6. Feb. 1897:174-178.
- Wright, Joseph. 1910. *Grammar of the Gothic Language*. Oxford. (Second edition with a supplement to the grammar by O.L. Sayce, 1954.)
- Zupitza, J. 1896. *Die germanischen Gutturale*. Berlin.

Note: I know of the existence of two unpublished studies on the Germanic numerals by the late Professors Ross and Tolkien respectively. As I have not been able to see them, I do not know if they contain anything of relevance to the history of the numeral '4'.